

Mathematical Methods · Medium paper

Real VCAA questions ranked by the published state results. Editable: type your answers straight into this PDF, save it, then mark yourself with the guide at the back.

Name _____ Date _____

Exam 1 ·

37 marks

reading 15 min · writing 60 min

Exam 2 ·

75 marks

reading 15 min · writing 120 min

State average on these questions

59 / 112 (52%)

a student at the state average on every question

How to use this paper

1. Open each question with its blue link. It opens the official VCAA exam PDF at the right page. The question text isn't reproduced here because it is VCAA copyright (some exams include third-party material).
2. Answer in this file: click a bubble for multiple choice, and type in the boxes for written answers (long answers scroll inside the box). Methods and science working can be typed in plain text, e.g. $v = \sqrt{2gh} = 9.9 \text{ m/s}$.
3. Keep to the real time limits above, and don't look at the guide until you've finished.
4. Save with Ctrl/Cmd + S. Use a PDF app that supports forms: Adobe Acrobat Reader, Chrome, Edge, Firefox, macOS Preview or Xodo.
5. Mark yourself: in the guide at the back, tick each marking point you earned, enter your marks, and compare your total with the state average.

Why these questions

Every question is a real VCAA exam question, chosen by ranking all the questions from the listed exams by how Victorian students actually scored on them (percentage correct for multiple choice, state average $\div$ marks for written questions). This paper takes the middle band of questions.

Sources

- [2023 Exam 1 Mathematical Methods exam \(PDF\)](#) · [examiners' report](#)
- [2024 Exam 1 Mathematical Methods exam \(PDF\)](#) · [examiners' report](#)
- [2025 Exam 1 Mathematical Methods exam \(PDF\)](#) · [examiners' report](#)
- [2023 Exam 2 Mathematical Methods exam \(PDF\)](#) · [examiners' report](#)
- [2024 Exam 2 Mathematical Methods exam \(PDF\)](#) · [examiners' report](#)
- [2025 Exam 2 Mathematical Methods exam \(PDF\)](#) · [examiners' report](#)

Exam 1 · technology free (37 marks, reading 15 min, writing 60 min)

Questions (technology free)

Open each question in the official exam PDF, then type your answer to every part in the boxes. Show working where more than one mark is available.

Question 1 (2 marks)

2023 VCAA Exam 1 · Question 4 · [open in the official exam PDF \(p. 5\)](#) ↗

Two trapeziums to approximate the area under $y = x + \frac{1}{x}$ from $x = 1$ to 3

Trapezium approximation (2 marks)

Question 2 (7 marks)

2023 VCAA Exam 1 · Question 7 · [open in the official exam PDF \(p. 8\)](#) ↗

$f(x) = x^2 - 2x$ on $(-\infty, 1]$: range, the inverse graph and rule, area enclosed with $y = -x$

a. Range of f (1 mark)

b. Sketch $y = f^{-1}(x)$ with endpoints and intercepts (2 marks)

c. Rule and domain of the inverse (2 marks)

d. Total area enclosed by f , f^{-1} and $y = -x$ (2 marks)

Question 3 (6 marks)

2023 VCAA Exam 1 · Question 9 · [open in the official exam PDF \(p. 12\)](#) ↗

Two walking tracks (a cubic and a quadratic): find the parameters, a shared turning point, and the maximum area of a triangle

a. Verify $a = 12$ and $b = -3$ (1 mark)

b. Verify both have a turning point at P and give P (2 marks)

c. Maximum area of triangle OAB with A(k, 0), B(k, g(k)) (3 marks)

Question 4 (3 marks)

2024 VCAA Exam 1 · Question 2 · [open in the official exam PDF \(p. 3\)](#) ↗

Value of k for which a pair of simultaneous linear equations has no solution

Find k for no solution (3 marks)

Question 5 (5 marks)

2024 VCAA Exam 1 · Question 3 · [open in the official exam PDF \(p. 4\)](#) ↗

Sketch the truncus $g(x) = \frac{1}{(x+3)^2} - 2$; area between it and the x-axis from $x = -2$ to 0

a. Sketch with asymptotes and intercepts (3 marks)

b. Area bounded by $x = -2$, the axes and the graph (2 marks)

Question 6 (3 marks)

2024 VCAA Exam 1 · Question 4 · [open in the official exam PDF \(p. 5\)](#) ↗

$X \sim \text{Bi}\left(4, \frac{9}{10}\right)$: standard deviation and $\Pr(X < 2)$

a. Standard deviation (1 mark)

b. $\Pr(X < 2)$ (2 marks)

Question 7 (6 marks)

2024 VCAA Exam 1 · Question 5 · [open in the official exam PDF \(p. 6\)](#) ↗

Population model $h(t) = \frac{3000}{t+1}$: transformations to halve the rate; a confidence interval and sample size

a. Population after four years (1 mark)

b. Two transformations taking h to h_1 with $h_1(0) = h(0)$ and half the rate of decrease (2 marks)

c.i. Approximate 95% CI with $\hat{p} = 0.6$, $n = 100$, $z = 2$ (2 marks)

c.ii. Smallest n giving a standard deviation of $\frac{\sqrt{2}}{50}$ (1 mark)

Question 8 (2 marks)

2025 VCAA Exam 1 · Question 2 · [open in the official exam PDF \(p. 3\)](#) ↗

Find $g(x)$ given $g'(x) = \frac{1}{2x+3}$ and $g(1) = 0$

Antidifferentiate and use the condition (2 marks)

Question 9 (3 marks)

2025 VCAA Exam 1 · Question 6 · [open in the official exam PDF \(p. 7\)](#) ↗

$X \sim \text{Bi}\left(6, \frac{1}{4}\right)$: variance and $\Pr(X \geq 5)$

a. $\text{var}(X)$ (1 mark)

b. $\Pr(X \geq 5)$ in the form $\frac{a}{2^b}$ (2 marks)

Exam 2 · CAS and a bound reference allowed (75 marks, reading 15 min, writing 120 min)

Section A – Multiple-choice questions

Open each question in the official exam PDF (click the blue link), then select A–D. One mark each.

Question 1 · 1 mark

2023 VCAA Exam 2 · Section A · Question 2 · [open in the official exam PDF \(p. 2\)](#) ↗

Axis of symmetry of $y = ax^2 + 2bx + c$

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 2 · 1 mark

2023 VCAA Exam 2 · Section A · Question 3 · [open in the official exam PDF \(p. 3\)](#) ↗

Domain of the sum of two functions with domains $[-2, 3]$ and $(-1, 5]$

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 3 · 1 mark

2023 VCAA Exam 2 · Section A · Question 4 · [open in the official exam PDF \(p. 3\)](#) ↗

Values of k giving infinitely many solutions

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 4 · 1 mark

2023 VCAA Exam 2 · Section A · Question 6 · [open in the official exam PDF \(p. 4\)](#) ↗

$\int_7^3 f \, dx$ from $\int_3^{10} f = C$ and $\int_7^{10} f = D$

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 5 · 1 mark

2023 VCAA Exam 2 · Section A · Question 8 · [open in the official exam PDF \(p. 4\)](#) ↗

Probability of at least one green in 8 draws with replacement

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 6 · 1 mark

2023 VCAA Exam 2 · Section A · Question 13 · [open in the official exam PDF \(p. 6\)](#) ↗

Result of three Newton's method iterations on $x^3 + 3x - 3$ from $x_0 = 1$

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 7 · 1 mark

2024 VCAA Exam 2 · Section A · Question 5 · [open in the official exam PDF \(p. 3\)](#) ↗

Range of $g(f(x))$ with $f(x) = x^2 - 4x$ on $(1, \infty)$ and $g(x) = e^{-x}$

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 8 · 1 mark

2024 VCAA Exam 2 · Section A · Question 6 · [open in the official exam PDF \(p. 3\)](#) ↗

Inverse of $f(x) = \frac{2x+1}{3-x}$ and its domain

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 9 · 1 mark2024 VCAA Exam 2 · Section A · Question 12 · [open in the official exam PDF \(p. 6\)](#) ↗Graph of $y = f(2x + 1)$ from the graph of f Your answer: ☐ A ☐ B ☐ C ☐ D

Question 10 · 1 mark2024 VCAA Exam 2 · Section A · Question 14 · [open in the official exam PDF \(p. 7\)](#) ↗ $\Pr(X < 0.5)$ for a piecewise-linear pdf with unknown k Your answer: ☐ A ☐ B ☐ C ☐ D

Question 11 · 1 mark2024 VCAA Exam 2 · Section A · Question 15 · [open in the official exam PDF \(p. 8\)](#) ↗Points of inflection of $y = 2 - \tan\left(\pi\left(x - \frac{1}{4}\right)\right)$ Your answer: ☐ A ☐ B ☐ C ☐ D

Question 12 · 1 mark2024 VCAA Exam 2 · Section A · Question 18 · [open in the official exam PDF \(p. 10\)](#) ↗Value of x maximising the area of a trapezium with slant sides 10Your answer: ☐ A ☐ B ☐ C ☐ D

Question 13 · 1 mark2024 VCAA Exam 2 · Section A · Question 19 · [open in the official exam PDF \(p. 10\)](#) ↗ $\Pr(X < 12)$ given $\Pr(X < 10) = \Pr(X > 18) = 0.2$ Your answer: ☐ A ☐ B ☐ C ☐ D

Question 14 · 1 mark2025 VCAA Exam 2 · Section A · Question 2 · [open in the official exam PDF \(p. 2\)](#) ↗All asymptotes of $y = 2 \tan\left(\pi\left(x + \frac{1}{2}\right)\right)$ Your answer: ☐ A ☐ B ☐ C ☐ D

Question 15 · 1 mark2025 VCAA Exam 2 · Section A · Question 4 · [open in the official exam PDF \(p. 4\)](#) ↗Values of k for which a 2×2 linear system has no solutionYour answer: ☐ A ☐ B ☐ C ☐ D

Question 16 · 1 mark2025 VCAA Exam 2 · Section A · Question 6 · [open in the official exam PDF \(p. 4\)](#) ↗For which function the trapezium rule overestimates the area on $[0, 1]$ Your answer: ☐ A ☐ B ☐ C ☐ D

Question 17 · 1 mark2025 VCAA Exam 2 · Section A · Question 9 · [open in the official exam PDF \(p. 6\)](#) ↗

Conditional probability that a student walked, given they took at least 30 minutes

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 18 · 1 mark2025 VCAA Exam 2 · Section A · Question 10 · [open in the official exam PDF \(p. 6\)](#) ↗Solving $(f \circ g)(x) > 0$ for $f(x) = 2x^2 + x - 1$, $g(x) = \sin x$ Your answer: ☐ A ☐ B ☐ C ☐ D

Question 19 · 1 mark

2025 VCAA Exam 2 · Section A · Question 12 · [open in the official exam PDF \(p. 7\)](#) ↗

Finding μ and σ from two normal probabilities

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 20 · 1 mark

2025 VCAA Exam 2 · Section A · Question 18 · [open in the official exam PDF \(p. 11\)](#) ↗

Which two probability mass functions have the same mean

Your answer: ☐ A ☐ B ☐ C ☐ D

Section B – Extended-response questions

Open each question in the official exam PDF, then type your answer to every part in the boxes. Show working where more than one mark is available.

Question 1 (15 marks)

2023 VCAA Exam 2 · Section B · Question 4 · [open in the official exam PDF \(p. 18\)](#) ↗

Tennis-ball diameters (normal), fitting through a container, grading, a confidence level, and serving-speed pdfs

a. $\Pr(D > 6.8)$ (4 d.p.) (1 mark)

b. Diameter exceeded by only 10% of balls (2 d.p.) (1 mark)

c. $\Pr(\text{ball fits: } D < 6.95)$ (4 d.p.) (1 mark)

d. $\Pr(\text{at least 3 of 4 fit})$ (4 d.p.) (2 marks)

e. $\Pr(\text{grade A} \mid \text{fits})$ (4 d.p.) (2 marks)

f. σ so that more than 99% are grade A (2 d.p.) (2 marks)

g. Confidence level of the interval (0.7382, 0.9493) with $n = 32$ (2 marks)

h. $\Pr(\text{grade A serving speed} > 50)$ (4 d.p.) (1 mark)

i. Exact mean serving speed of grade A balls (1 mark)

j. a and b in $g(w) = af\left(\frac{w}{b}\right)$ if the mean is $2\pi^2 + 8$ (2 marks)

Question 2 (11 marks)

2024 VCAA Exam 2 · Section B · Question 3 · [open in the official exam PDF \(p. 18\)](#) ↗

Modelling monthly online sales: a cubic through turning points, translating it, then a trend-plus-cosines model

a.i. a, b, c, d for the cubic with a min at (2, 2500) and max at (11, 4400) (2 d.p.) (3 marks)

a.ii. h and k so that $q(t) = p(t - h) + k$ has a local max at $(23, 4750)$ (2 marks)

b.i. Complete the graph of f to $t = 36$ and label the endpoint (2 marks)

b.ii. Yearly increase n predicted by f (1 mark)

b.iii. $f'(t)$ (1 mark)

b.iv. Maximum instantaneous rate of change and when it occurs (2 marks)

Question 3 (15 marks)

2024 VCAA Exam 2 · Section B · Question 4 · [open in the official exam PDF \(p. 21\)](#) ↗

Luggage masses (a cubic pdf), a discrete count of heavy bags per traveller, sample proportions and confidence intervals

a. Integral for $\Pr(\text{heavy})$, heavy meaning > 23 kg (1 mark)

b.i. Mean of X (1 mark)

b.ii. Standard deviation of X (2 marks)

b.iii. $\Pr(\text{heavy} \mid \text{more than the mean})$ (3 d.p.) (2 marks)

c.i. Show $\Pr(W = 2) = 0.027$ (1 mark)

c.ii. Complete the distribution of W (3 d.p.) (2 marks)

d.i. $\Pr(\hat{P} > 0.2)$ for $n = 35$ (3 d.p.) (2 marks)

d.ii. $\Pr(\hat{P} \text{ within one sd of its mean}), \text{ exact binomial}$ (3 d.p.) (2 marks)

e.i. Approximate 90% CI from 10 heavy out of 50 (3 d.p.) (1 mark)

e.ii. Min and max number of heavy bags in a second sample of 50 giving a wider CI (1 mark)

Question 4 (14 marks)

2025 VCAA Exam 2 · Section B · Question 2 · [open in the official exam PDF \(p. 16\)](#) ↗

A line and an exponential meeting at $(-12, 1)$ and $(2, 8)$: find A and k, area between them, the inverse, antiderivatives

a. Show $A = 2^{18/7}$ and $k = \frac{3}{14} \log_e 2$ (3 marks)

b. b with $g(x) = A \times 2^{bx}$ (1 mark)

c. Area between the graphs on $[-12, 2]$ (2 d.p.) (2 marks)

d.i. Expression for $h'(x)$, $h = f - g$ (1 mark)

d.ii. Maximum of h on $[-12, 2]$ (2 d.p.) (1 mark)

e. Where $y = g^{-1}(x)$ meets $y = 2(x - 7)$ (2 marks)

f.i. Show F (antiderivative of f through $(0, c)$) can't pass through both $(-12, 1)$ and $(2, 8)$ (2 marks)

f.ii. Dilation factor m and c so that mF passes through both points (2 marks)

END OF PAPER

Marking guide follows

Stop here until you have finished the paper under exam conditions.

Marking guide · Exam 1

Questions (technology free)

Question 1 (2 marks) · state average 1.1 / 2 (55%)

2023 VCAA Exam 1 · Question 4

Trapezium approximation

☐ Values $f(1) = 2$, $f(2) = \frac{5}{2}$, $f(3) = \frac{10}{3}$

☐ Area $\approx \frac{1}{2} \left(2 + 2 \times \frac{5}{2} + \frac{10}{3} \right) = \frac{31}{6}$

Answer / notes: $\frac{31}{6}$

State average 1.1 / 2 (55%); 45% scored full marks. Mark distribution: 0: 35% · 1: 20% · 2: 45%

Your mark: _____ / 2

Question 2 (7 marks) · state average 3.5 / 7 (50%)

2023 VCAA Exam 1 · Question 7

a. Range of f

☐ $[-1, \infty)$

Answer / notes: $[-1, \infty)$

State average 0.9 / 1 (90%); 90% scored full marks. Mark distribution: 0: 10% · 1: 90%

Your mark: _____ / 1

b. Sketch $y = f^{-1}(x)$ with endpoints and intercepts

☐ Reflection of f in $y = x$, starting from the endpoint $(-1, 1)$

☐ Passes through $(0, 0)$, decreasing

Answer / notes: Closed endpoint $(-1, 1)$, through the origin, heading down to the right.

State average 1.1 / 2 (55%); 47% scored full marks. Mark distribution: 0: 33% · 1: 20% · 2: 47%

Your mark: _____ / 2

c. Rule and domain of the inverse

☐ $f^{-1}(x) = 1 - \sqrt{x+1}$

☐ Domain $[-1, \infty)$

Answer / notes: Complete the square: $y = 1 - \sqrt{x+1}$ (negative root because $y \leq 1$).

State average 1.0 / 2 (50%); 21% scored full marks. Mark distribution: 0: 26% · 1: 54% · 2: 21%

Your mark: _____ / 2

d. Total area enclosed by f , f^{-1} and $y = -x$

☐ Area between f and $y = -x$ on $[0, 1]$: $\int_0^1 (x - x^2) dx = \frac{1}{6}$

☐ Doubled by symmetry about $y = x$: $\frac{1}{3}$

Answer / notes: $\frac{1}{6} + \frac{1}{6} = \frac{1}{3}$

State average 0.5 / 2 (25%); 19% scored full marks. Mark distribution: 0: 71% · 1: 10% · 2: 19%

Your mark: _____ / 2

Question 3 (6 marks) · state average 2.7 / 6 (45%)

2023 VCAA Exam 1 · Question 9

a. Verify $a = 12$ and $b = -3$

☐ $f(0) = a = 12$ and $g(1) = 12 + b = 9 \Rightarrow b = -3$

Answer / notes: Direct substitution.

State average 0.9 / 1 (90%); 90% scored full marks. Mark distribution: 0: 10% · 1: 90%

Your mark: _____ / 1

b. Verify both have a turning point at P and give P

☐ $f'(x) = -(x-2)(3x-2)$ and $g'(x) = 12 - 6x$, both zero at $x = 2$

☐ $P = (2, 12)$

Answer / notes: Both derivatives vanish at $x = 2$, and $f(2) = g(2) = 12$.

State average 1.0 / 2 (50%); 30% scored full marks. Mark distribution: 0: 34% · 1: 36% · 2: 30%

Your mark: _____ / 2

c. Maximum area of triangle OAB with A(k, 0), B(k, g(k))

☐ $A(k) = \frac{1}{2}k(12k - 3k^2) = 6k^2 - \frac{3}{2}k^3$

☐ $A'(k) = 12k - \frac{9}{2}k^2 = 0 \Rightarrow k = \frac{8}{3}$

☐ Maximum area $\frac{128}{9} \text{ km}^2$

Answer / notes: $k = \frac{8}{3}$ gives $\frac{128}{9} \text{ km}^2$.

State average 0.8 / 3 (27%); 13% scored full marks. Mark distribution: 0: 65% · 1: 5% · 2: 17% · 3: 13%

Your mark: _____ / 3

Question 4 (3 marks) · state average 1.7 / 3 (57%)

2024 VCAA Exam 1 · Question 2

Find k for no solution

☐ Sets the determinant (or gradient ratio) to zero: $3k^2 + 2k - 8 = 0$

☐ $k = \frac{4}{3}$ or $k = -2$

☐ Checks: $k = -2$ gives identical lines (infinitely many), so $k = \frac{4}{3}$

Answer / notes: $k = \frac{4}{3}$. You must reject $k = -2$, which gives the same line twice.

State average 1.7 / 3 (57%); 37% scored full marks. Mark distribution: 0: 22% · 1: 22% · 2: 18% · 3: 37%

Your mark: _____ / 3

Question 5 (5 marks) · state average 2.9 / 5 (58%)

2024 VCAA Exam 1 · Question 3

a. Sketch with asymptotes and intercepts

☐ Asymptotes $x = -3$ and $y = -2$

☐ y-intercept $(0, -\frac{17}{9})$

☐ x-intercepts $(-3 \pm \frac{\sqrt{2}}{2}, 0)$, symmetric about $x = -3$

Answer / notes: Truncus above $y = -2$, symmetric about $x = -3$, crossing the x-axis at $-3 \pm \frac{1}{\sqrt{2}}$.

State average 2.1 / 3 (70%); 42% scored full marks. Mark distribution: 0: 8% · 1: 16% · 2: 35% · 3: 42%

Your mark: _____ / 3

b. Area bounded by $x = -2$, the axes and the graph

☐ Sets up $-\int_{-2}^0 \left(\frac{1}{(x+3)^2} - 2 \right) dx$ (region below the axis)

☐ Area = $\frac{10}{3}$

Answer / notes: $\int_{-2}^0 g dx = \frac{2}{3} - 4 = -\frac{10}{3}$, so the area is $\frac{10}{3}$.

State average 0.8 / 2 (40%); 27% scored full marks. Mark distribution: 0: 48% · 1: 25% · 2: 27%

Your mark: _____ / 2

Question 6 (3 marks) · state average 1.6 / 3 (53%)

2024 VCAA Exam 1 · Question 4

a. Standard deviation

☐ $\sqrt{4 \times 0.9 \times 0.1} = 0.6$

Answer / notes: 0.6. Many gave the variance (0.36) instead.

State average 0.6 / 1 (60%); 59% scored full marks. Mark distribution: 0: 41% · 1: 59%

Your mark: _____ / 1

b. $\Pr(X < 2)$

☐ $\Pr(X = 0) + \Pr(X = 1) = 0.1^4 + 4(0.9)(0.1)^3$

☐ $= 0.0037$

Answer / notes: $0.0001 + 0.0036 = 0.0037$. Watch the decimal places.

State average 1.0 / 2 (50%); 39% scored full marks. Mark distribution: 0: 38% · 1: 23% · 2: 39%

Your mark: _____ / 2

Question 7 (6 marks) · state average 3.1 / 6 (52%)

2024 VCAA Exam 1 · Question 5

a. Population after four years

☐ 600

Answer / notes: $h(4) = 600$

State average 1.0 / 1 (100%); 95% scored full marks. Mark distribution: 0: 5% · 1: 95%

Your mark: _____ / 1

b. Two transformations taking h to h_1 with $h_1(0) = h(0)$ and half the rate of decrease

☐ Dilation by a factor of $\frac{1}{2}$ from the t-axis

☐ Then a translation of 1500 up (or up 3000 then dilate by $\frac{1}{2}$)

Answer / notes: $h_1(t) = \frac{1}{2}h(t) + 1500$. Most students tried a horizontal dilation, which changes the timing, not the rate at each time.

State average 0.4 / 2 (20%); 2% scored full marks. Mark distribution: 0: 69% · 1: 29% · 2: 2%

Your mark: _____ / 2

c.i. Approximate 95% CI with $\hat{p} = 0.6$, $n = 100$, $z = 2$

☐ $\text{sd}(\hat{P}) = \sqrt{\frac{0.6 \times 0.4}{100}} = \frac{\sqrt{6}}{50}$

☐ $\text{CI} = \left(0.6 - \frac{\sqrt{6}}{25}, 0.6 + \frac{\sqrt{6}}{25}\right)$

Answer / notes: $0.6 \pm 2 \times \frac{\sqrt{6}}{50}$

State average 1.3 / 2 (65%); 44% scored full marks. Mark distribution: 0: 19% · 1: 37% · 2: 44%

Your mark: _____ / 2

c.ii. Smallest n giving a standard deviation of $\frac{\sqrt{2}}{50}$

☐ $n = 300$

Answer / notes: $\frac{0.24}{n} = \frac{2}{2500} \Rightarrow n = 300$

State average 0.4 / 1 (40%); 33% scored full marks. Mark distribution: 0: 67% · 1: 33%

Your mark: _____ / 1

Question 8 (2 marks) · state average 1.2 / 2 (60%)

2025 VCAA Exam 1 · Question 2

Antidifferentiate and use the condition

☐ $g(x) = \frac{1}{2} \log_e(2x + 3) + c$

☐ $c = -\frac{1}{2} \log_e 5$, so $g(x) = \frac{1}{2} \log_e\left(\frac{2x+3}{5}\right)$

Answer / notes: $g(x) = \frac{1}{2} \log_e(2x + 3) - \frac{1}{2} \log_e 5$. Don't forget the $\frac{1}{2}$ from the chain rule.

State average 1.2 / 2 (60%); 40% scored full marks. Mark distribution: 0: 22% · 1: 38% · 2: 40%

Your mark: _____ / 2

Question 9 (3 marks) · state average 1.8 / 3 (60%)

2025 VCAA Exam 1 · Question 6

a. $\text{var}(X)$

☐ $\frac{9}{8}$

Answer / notes: $np(1 - p) = 6 \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{9}{8}$

State average 0.8 / 1 (80%); 73% scored full marks. Mark distribution: 0: 27% · 1: 73%

Your mark: _____ / 1

b. $\Pr(X \geq 5)$ in the form $\frac{a}{2^b}$

☐ $\binom{6}{5} \left(\frac{1}{4}\right)^5 \frac{3}{4} + \left(\frac{1}{4}\right)^6$

☐ $\frac{19}{2^{12}}$

Answer / notes: $\frac{18+1}{4^6} = \frac{19}{2^{12}}$

State average 1.0 / 2 (50%); 34% scored full marks. Mark distribution: 0: 31% · 1: 35% · 2: 34%

Your mark: _____ / 2

Marking guide · Exam 2

Section A – Multiple-choice questions

✓	Q	Ans	Why	How Victoria went
☐	1	A	$x = -\frac{2b}{2a} = -\frac{b}{a}$. The coefficient of x is 2b, not b. <i>Most popular wrong answer: B (29%).</i>	53% correct A 53% · B 29% · C 3% · D 6% · E 8%
☐	2	E	The sum's domain is the intersection: $(-1, 3)$. <i>Most popular wrong answer: A (22%).</i>	47% correct A 22% · B 12% · C 11% · D 7% · E 47%
☐	3	B	The determinant $k^2 + k - 20 = 0$ gives $k = -5$ or 4. Only $k = -5$ makes the equations identical. $k = 4$ gives parallel lines. <i>Most popular wrong answer: A (15%).</i>	55% correct A 15% · B 55% · C 10% · D 14% · E 4%
☐	4	D	$\int_3^7 f = C - D$, and reversing the terminals gives $D - C$. <i>Most popular wrong answer: C (41%).</i>	49% correct A 3% · B 4% · C 41% · D 49% · E 2%
☐	5	C	$1 - \Pr(\text{no green}) = 1 - \left(\frac{m}{n+m}\right)^8$. <i>Most popular wrong answer: D (16%).</i>	49% correct A 12% · B 14% · C 49% · D 16% · E 9%
☐	6	C	$1 \rightarrow 0.83333 \rightarrow 0.81785 \rightarrow 0.81773$. Count exactly three iterations. <i>Most popular wrong answer: B (22%).</i>	52% correct A 16% · B 22% · C 52% · D 7% · E 3%
☐	7	D	f has range $[-4, \infty)$, so e^{-f} has range $(0, e^4]$. <i>Most popular wrong answer: C (21%).</i>	52% correct A 18% · B 9% · C 21% · D 52%
☐	8	B	Swap x and y and solve: $y = 3 - \frac{7}{x+2}$. The domain of f^{-1} is the range of f, which is $\mathbb{R} \setminus \{-2\}$. <i>Most popular wrong answer: A (31%).</i>	50% correct A 31% · B 50% · C 7% · D 12%
☐	9	A	Each point (x, y) maps to $(\frac{x-1}{2}, y)$: shift left 1, then halve the distance from the y-axis. Track the turning points. <i>Most popular wrong answer: B (28%).</i>	47% correct A 47% · B 28% · C 17% · D 7%
☐	10	B	The total area $4k - 1 = 1$ gives $k = \frac{1}{2}$. Then $\Pr(X < 0.5) = 1 - \frac{1}{16} = \frac{15}{16}$. <i>Most popular wrong answer: C (30%).</i>	53% correct A 9% · B 53% · C 30% · D 7%
☐	11	A	$\tan$ inflects where its argument is $k\pi$: $x = k + \frac{1}{4}$, $y = 2$. <i>Most popular wrong answer: B (19%).</i>	53% correct A 53% · B 19% · C 13% · D 15%
☐	12	B	$A = 2x\sqrt{100 - x^2}$. Maximise $x^2(100 - x^2)$: $x^2 = 50$, so $x = 5\sqrt{2}$. <i>Most popular wrong answer: A (18%).</i>	58% correct A 18% · B 58% · C 12% · D 10%
☐	13	C	By symmetry $\mu = 14$, and $\sigma = \frac{4}{0.8416} = 4.75$. Then $\Pr(X < 12) \approx 0.337$. <i>Most popular wrong answer: B (19%).</i>	57% correct A 10% · B 19% · C 57% · D 13%
☐	14	A	Set $\pi(x + \frac{1}{2}) = \frac{\pi}{2} + k\pi$, so $x = k$, $k \in \mathbb{Z}$. <i>Most popular wrong answer: C (21%).</i>	51% correct A 51% · B 10% · C 21% · D 18%
☐	15	A	The determinant $2k^2 + k - 6 = 0$ gives $k = -2$ or $\frac{3}{2}$. $k = \frac{3}{2}$ gives the same line twice (infinitely many solutions), so only $k = -2$. <i>Most popular wrong answer: C (19%).</i>	55% correct A 55% · B 13% · C 19% · D 12%

✓	Q	Ans	Why	How Victoria went
☐	16	B	The trapezium rule overestimates when the curve is concave up on the interval. $x^3 + 1$ is concave up on $[0, 1]$. <i>Most popular wrong answer: C (19%)</i> .	50% correct A 18% · B 50% · C 19% · D 11%
☐	17	A	$\frac{0.2m}{0.2m+0.4n} = \frac{m}{m+2n}$ <i>Most popular wrong answer: B (19%)</i> .	57% correct A 57% · B 19% · C 16% · D 7%
☐	18	C	$(2 \sin x - 1)(\sin x + 1) > 0$. Since $\sin x \geq -1$, you need $\frac{1}{2} < \sin x \leq 1$. <i>Most popular wrong answer: D (17%)</i> .	57% correct A 10% · B 15% · C 57% · D 17%
☐	19	D	$\Pr(X > 200) = 0.325 \Rightarrow z = 0.454$, and $\Pr(X < 180) = 0.086 \Rightarrow z = -1.366$. Solve simultaneously: $\mu \approx 195$, $\sigma \approx 11$. <i>Most popular wrong answer: B (16%)</i> .	58% correct A 10% · B 16% · C 15% · D 58%
☐	20	D	Compute each mean: 2.5, 3.0, 2.0 and 3.0. The two with mean 3 match. <i>Most popular wrong answer: B (17%)</i> .	50% correct A 16% · B 17% · C 17% · D 50%

Multiple-choice score: / 20

Section B – Extended-response questions

Question 1 (15 marks) · state average 7.3 / 15 (49%)

2023 VCAA Exam 2 · Section B · Question 4

a. $\Pr(D > 6.8)$ (4 d.p.)

☐ ≈ 0.1587

Answer / notes: ≈ 0.1587

State average 0.8 / 1 (80%); 79% scored full marks. Mark distribution: 0: 21% · 1: 79%

Your mark: / 1

b. Diameter exceeded by only 10% of balls (2 d.p.)

☐ ≈ 6.83 cm

Answer / notes: 90th percentile ≈ 6.83 cm

State average 0.6 / 1 (60%); 59% scored full marks. Mark distribution: 0: 41% · 1: 59%

Your mark: / 1

c. $\Pr(\text{ball fits: } D < 6.95)$ (4 d.p.)

☐ ≈ 0.9938

Answer / notes: ≈ 0.9938

State average 0.8 / 1 (80%); 77% scored full marks. Mark distribution: 0: 23% · 1: 77%

Your mark: / 1

d. $\Pr(\text{at least 3 of 4 fit})$ (4 d.p.)

☐ Binomial with $n = 4$, $p = 0.9938$

☐ ≈ 0.9998

Answer / notes: ≈ 0.9998

State average 1.3 / 2 (65%); 54% scored full marks. Mark distribution: 0: 28% · 1: 18% · 2: 54%

Your mark: / 2

e. $\Pr(\text{grade A} \mid \text{fits})$ (4 d.p.)

☐ $\frac{\Pr(6.54 < D < 6.86)}{\Pr(D < 6.95)}$

☐ ≈ 0.8960

Answer / notes: ≈ 0.8960

State average 1.2 / 2 (60%); 53% scored full marks. Mark distribution: 0: 31% · 1: 16% · 2: 53%

Your mark: / 2

f. σ so that more than 99% are grade A (2 d.p.)

☐ $\frac{0.16}{\sigma} = 2.5758$

☐ $\sigma \approx 0.06 \text{ cm}$

Answer / notes: Symmetric interval, so 0.5% in each tail.

State average 0.7 / 2 (35%); 26% scored full marks. Mark distribution: 0: 59% · 1: 15% · 2: 26%

Your mark: _____ / 2

g. Confidence level of the interval (0.7382, 0.9493) with $n = 32$

☐ $\hat{p} = 0.84375$, margin 0.10555, so $z \approx 1.645$

☐ 90%

Answer / notes: 90%. Many found z correctly but then said 95%.

State average 0.5 / 2 (25%); 19% scored full marks. Mark distribution: 0: 66% · 1: 15% · 2: 19%

Your mark: _____ / 2

h. $\Pr(\text{grade A serving speed} > 50)$ (4 d.p.)

☐ ≈ 0.1345

Answer / notes: $\int_{50}^{3\pi^2+30} f(v) dv \approx 0.1345$

State average 0.6 / 1 (60%); 57% scored full marks. Mark distribution: 0: 43% · 1: 57%

Your mark: _____ / 1

i. Exact mean serving speed of grade A balls

☐ $3\pi^2 + 12$

Answer / notes: $E(V) = 3\pi^2 + 12$

State average 0.6 / 1 (60%); 55% scored full marks. Mark distribution: 0: 45% · 1: 55%

Your mark: _____ / 1

j. a and b in $g(w) = af\left(\frac{w}{b}\right)$ if the mean is $2\pi^2 + 8$

☐ Area stays 1, so $ab = 1$; the mean scales by b : $b(3\pi^2 + 12) = 2\pi^2 + 8$

☐ $b = \frac{2}{3}$, $a = \frac{3}{2}$

Answer / notes: $b = \frac{2}{3}$, $a = \frac{3}{2}$

State average 0.2 / 2 (10%); 6% scored full marks. Mark distribution: 0: 89% · 1: 5% · 2: 6%

Your mark: _____ / 2

Question 2 (11 marks) · state average 5.8 / 11 (53%)

2024 VCAA Exam 2 · Section B · Question 3

a.i. a, b, c, d for the cubic with a min at (2, 2500) and max at (11, 4400) (2 d.p.)

☐ Four equations: $p(2) = 2500$, $p(11) = 4400$, $p'(2) = 0$, $p'(11) = 0$

☐ Solves the system with CAS

☐ $a \approx -5.21$, $b \approx 101.65$, $c \approx -344.03$, $d \approx 2823.18$

Answer / notes: Set up all four conditions, then solve on CAS.

State average 1.7 / 3 (57%); 42% scored full marks. Mark distribution: 0: 30% · 1: 16% · 2: 12% · 3: 42%

Your mark: _____ / 3

a.ii. h and k so that $q(t) = p(t - h) + k$ has a local max at (23, 4750)

☐ $h = 12$

☐ $k = 350$

Answer / notes: Move (11, 4400) to (23, 4750).

State average 0.6 / 2 (30%); 26% scored full marks. Mark distribution: 0: 65% · 1: 9% · 2: 26%

Your mark: _____ / 2

b.i. Complete the graph of f to $t = 36$ and label the endpoint

- ☐ Correct shape continuing the pattern (rising trend with a 12-month cycle)
- ☐ Endpoint $(36, 5180)$

Answer / notes: $f(36) = 3000 + 1080 + 700 + 400 = 5180$.

State average $1.5 / 2$ (75%); 63% scored full marks. Mark distribution: 0: 19% · 1: 18% · 2: 63%

Your mark: _____ / 2

b.ii. Yearly increase n predicted by f

- ☐ $n = 360$

Answer / notes: The cosines repeat every 12 months, so $f(t + 12) - f(t) = 30 \times 12 = 360$.

State average $0.6 / 1$ (60%); 59% scored full marks. Mark distribution: 0: 41% · 1: 59%

Your mark: _____ / 1

b.iii. $f'(t)$

- ☐ $30 - \frac{700\pi}{6} \sin\left(\frac{\pi t}{6}\right) - \frac{400\pi}{3} \sin\left(\frac{\pi t}{3}\right)$

Answer / notes: Chain rule on each cosine.

State average $0.8 / 1$ (80%); 77% scored full marks. Mark distribution: 0: 23% · 1: 77%

Your mark: _____ / 1

b.iv. Maximum instantaneous rate of change and when it occurs

- ☐ ≈ 725 million dollars per month
- ☐ At $t \approx 10.2, 22.2, 34.2$

Answer / notes: Maximise f' on CAS: 725 at $t \approx 10.2$, and it repeats every 12 months.

State average $0.6 / 2$ (30%); 21% scored full marks. Mark distribution: 0: 65% · 1: 14% · 2: 21%

Your mark: _____ / 2

Question 3 (15 marks) · state average $7.2 / 15$ (48%)

2024 VCAA Exam 2 · Section B · Question 4

a. Integral for $\Pr(\text{heavy})$, heavy meaning > 23 kg

- ☐ $\int_{23}^{30} \frac{1}{67500} x^2 (30 - x) dx$

Answer / notes: $\Pr(X > 23) = \int_{23}^{30} f(x) dx \approx 0.234$

State average $0.8 / 1$ (80%); 79% scored full marks. Mark distribution: 0: 21% · 1: 79%

Your mark: _____ / 1

b.i. Mean of X

- ☐ 18 kg

Answer / notes: 18 kg

State average $0.9 / 1$ (90%); 85% scored full marks. Mark distribution: 0: 15% · 1: 85%

Your mark: _____ / 1

b.ii. Standard deviation of X

- ☐ Uses $E(X^2) - [E(X)]^2$ (var = 36)
- ☐ sd = 6 kg

Answer / notes: 6 kg

State average $1.5 / 2$ (75%); 71% scored full marks. Mark distribution: 0: 21% · 1: 9% · 2: 71%

Your mark: _____ / 2

b.iii. $\Pr(\text{heavy} \mid \text{more than the mean})$ (3 d.p.)

- ☐ $\frac{\Pr(X > 23)}{\Pr(X > 18)}$
- ☐ ≈ 0.446

Answer / notes: ≈ 0.446

State average $1.2 / 2$ (60%); 52% scored full marks. Mark distribution: 0: 31% · 1: 18% · 2: 52%

Your mark: _____ / 2

c.i. Show $\Pr(W = 2) = 0.027$

☐ $0.5 \times 0.234^2 \approx 0.027$

Answer / notes: Only travellers with two bags can have $W = 2$.

State average $0.4 / 1$ (40%); 36% scored full marks. Mark distribution: 0: 64% · 1: 36%

Your mark: _____ / 1

c.ii. Complete the distribution of W (3 d.p.)

☐ $\Pr(W = 0) = 0.1 + 0.4(0.766) + 0.5(0.766)^2 \approx 0.700$

☐ $\Pr(W = 1) = 0.4(0.234) + 0.5 \times 2(0.234)(0.766) \approx 0.273$

Answer / notes: Condition on the number of bags checked.

State average $0.3 / 2$ (15%); 11% scored full marks. Mark distribution: 0: 86% · 1: 3% · 2: 11%

Your mark: _____ / 2

d.i. $\Pr(\hat{P} > 0.2)$ for $n = 35$ (3 d.p.)

☐ Equivalent to $\Pr(X \geq 8)$, $X \sim \text{Bi}(35, 0.234)$

☐ ≈ 0.595

Answer / notes: $\hat{P} > 0.2 \Leftrightarrow X > 7. \approx 0.595$

State average $0.9 / 2$ (45%); 33% scored full marks. Mark distribution: 0: 50% · 1: 18% · 2: 33%

Your mark: _____ / 2

d.ii. $\Pr(\hat{P}$ within one sd of its mean), exact binomial (3 d.p.)

☐ $\text{sd}(\hat{P}) = \sqrt{\frac{0.234 \times 0.766}{35}}$ gives X between 5.69 and 10.69, so $X = 6, \dots, 10$

☐ ≈ 0.684

Answer / notes: Convert the interval into counts, then add the binomial probabilities: ≈ 0.684 .

State average $0.5 / 2$ (25%); 18% scored full marks. Mark distribution: 0: 67% · 1: 15% · 2: 18%

Your mark: _____ / 2

e.i. Approximate 90% CI from 10 heavy out of 50 (3 d.p.)

☐ (0.107, 0.293)

Answer / notes: $0.2 \pm 1.645 \sqrt{\frac{0.2 \times 0.8}{50}}$

State average $0.6 / 1$ (60%); 56% scored full marks. Mark distribution: 0: 44% · 1: 56%

Your mark: _____ / 1

e.ii. Min and max number of heavy bags in a second sample of 50 giving a wider CI

☐ 11 and 39

Answer / notes: A wider CI needs a larger $\hat{p}(1 - \hat{p})$, so $\hat{p}$ closer to 0.5 than 0.2 is: 11 to 39. Only 10% got this.

State average $0.1 / 1$ (10%); 8% scored full marks. Mark distribution: 0: 92% · 1: 8%

Your mark: _____ / 1

Question 4 (14 marks) · state average 8.1 / 14 (58%)

2025 VCAA Exam 2 · Section B · Question 2

a. Show $A = 2^{18/7}$ and $k = \frac{3}{14} \log_e 2$

☐ $Ae^{-12k} = 1$ and $Ae^{2k} = 8$

☐ Dividing: $e^{14k} = 8 \Rightarrow k = \frac{3}{14} \log_e 2$

☐ $A = e^{12k} = 2^{36/14} = 2^{18/7}$

Answer / notes: Divide the equations to eliminate A , then substitute back.

State average $1.5 / 3$ (50%); 31% scored full marks. Mark distribution: 0: 22% · 1: 35% · 2: 12% · 3: 31%

Your mark: _____ / 3

b. b with $g(x) = A \times 2^{bx}$

☐ $b = \frac{3}{14}$

Answer / notes: $e^{kx} = 2^{3x/14}$, so $b = \frac{3}{14}$.

State average $0.6 / 1$ (60%); 55% scored full marks. Mark distribution: 0: 45% · 1: 55%

Your mark: _____ / 1

c. Area between the graphs on $[-12, 2]$ (2 d.p.)

☐ $\int_{-12}^2 (f(x) - g(x)) dx$

☐ ≈ 15.87

Answer / notes: 15.87 square units. The line is above the exponential between the intersections.

State average 1.6 / 2 (80%); 77% scored full marks. Mark distribution: 0: 13% · 1: 10% · 2: 77%

Your mark: _____ / 2

d.i. Expression for $h'(x)$, $h = f - g$

☐ $h'(x) = \frac{1}{2} - Ake^{kx}$

Answer / notes: $\frac{1}{2} - \frac{3 \log_e 2}{14} 2^{18/7} e^{\frac{3 \log_e 2}{14} x}$

State average 0.7 / 1 (70%); 71% scored full marks. Mark distribution: 0: 29% · 1: 71%

Your mark: _____ / 1

d.ii. Maximum of h on $[-12, 2]$ (2 d.p.)

☐ ≈ 1.72

Answer / notes: At $x \approx -3.83$, $h \approx 1.72$.

State average 0.6 / 1 (60%); 56% scored full marks. Mark distribution: 0: 44% · 1: 56%

Your mark: _____ / 1

e. Where $y = g^{-1}(x)$ meets $y = 2(x - 7)$

☐ Recognises $2(x - 7) = f^{-1}(x)$, so reflect the intersections in $y = x$

☐ $(1, -12)$ and $(8, 2)$

Answer / notes: Reflect $(-12, 1)$ and $(2, 8)$ in $y = x$: $(1, -12)$ and $(8, 2)$.

State average 1.4 / 2 (70%); 62% scored full marks. Mark distribution: 0: 26% · 1: 12% · 2: 62%

Your mark: _____ / 2

f.i. Show F (antiderivative of f through $(0, c)$) can't pass through both $(-12, 1)$ and $(2, 8)$

☐ $F(x) = \frac{x^2}{4} + 7x + c$

☐ Requires $c = 49$ from one point and $c = -7$ from the other, a contradiction

Answer / notes: $F(-12) = c - 48$ and $F(2) = c + 15$. No single c fits both.

State average 1.0 / 2 (50%); 39% scored full marks. Mark distribution: 0: 41% · 1: 21% · 2: 39%

Your mark: _____ / 2

f.ii. Dilation factor m and c so that mF passes through both points

☐ $m(c - 48) = 1$ and $m(c + 15) = 8$

☐ $c = 57$, $m = \frac{1}{9}$

Answer / notes: Divide: $c + 15 = 8(c - 48) \Rightarrow c = 57$, then $m = \frac{1}{9}$.

State average 0.7 / 2 (35%); 28% scored full marks. Mark distribution: 0: 62% · 1: 11% · 2: 28%

Your mark: _____ / 2

Your result

Total: _____ / 112 State average on these questions: **59 / 112 (52%)**

Topics to revise:

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